

Average Payoff, Typical Ticket

Unit 2 · Topic 2.8 · THE PROBLEM

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	$1/200$
Second prize	23	$4/200$
Small prize	3	$20/200$
No prize	-2	$175/200$

FIRST TAKE — one sentence before you work the parts

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

- DOK 1** (a) State the four possible net profits X . Which is most likely?
- DOK 2** (b) Check that the listed probabilities add to 1. Find the chance that a ticket loses money.
- DOK 3** (c) Can Priya's average payout and Mateo's statement both be true? Use the distribution to explain. Decide whether Priya's good-deal claim follows for a buyer trying to make money, using the ticket cost and possible net profits. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.